DataPilot AI - Data Science Report

1. Dataset Overview

File Name	employee_attrition.csv
Rows	20
Columns	11

2. Dataset Understanding

Summary: The dataset contains 20 rows and 11 columns, with 7 categorical columns and 4 numeric columns. The data quality is good as there are no missing or duplicate values.

Problem Type: ProblemType.CLASSIFICATION

Observations:

- Attrition is a categorical variable that could be the target for predicting employee turnover.

- The dataset size is very small, which may limit the model's performance and generalization ability.

- Age, Salary, YearsAtCompany are potential features related to attrition.

3. Data Quality

Metric	Value
Duplicate Rows	0
Duplicate Percentage	0.00%
Constant Columns	0
Empty Columns	0

4. Model Evaluation

Metric	Value
accuracy	1.0000
precision	1.0000
recall	1.0000
f1_score	1.0000

5. Model Explainability

Top Features:

- Age: 0.6065

- YearsAtCompany: 0.5068

- MaritalStatus_Single: 0.4223

- Salary: 0.3600

- Education_Bachelor: 0.2368

- MaritalStatus_Married: 0.2364

- OverTime_Yes: 0.2204

- OverTime_No: 0.2201

- Education_Master: 0.2037

- Department_HR: 0.0999

6. Business Insights

Executive Summary

The dataset is small with only 20 rows, but it has a good data quality profile as there are no missing or duplicate values. The model predicts employee attrition accurately, achieving perfect precision, recall, and F1 score, based on the confusion matrix and classification report. Key features include age, years at company, marital status, salary, and department. However, due to the small dataset size, the generalizability of these results is limited, and further validation using a larger or synthetic dataset might be necessary.

Key Findings

- The model's accuracy, precision, recall, and F1 score are all 100%, suggesting that it perfectly predicts employee attrition based on the provided data.

- Age, YearsAtCompany, MaritalStatus_Single, Salary, and Department_HR are identified as significant features contributing to the prediction of employee attrition.

Recommendations

- Further validate the model using a larger dataset to ensure its generalizability and reliability in real-world scenarios.

- Consider conducting domain-specific feature engineering, such as including performance metrics or company culture factors, which can provide deeper insights into employee attrition.

- Explore implementing cross-validation techniques to better assess the model's robustness given the limited training data.

Risks and Limitations

- Due to the extremely small dataset size (only 20 samples), there is a risk of overfitting, which could lead to poor generalization on new or unseen data.

- The model might not capture all relevant factors affecting employee attrition due to the limited feature space. Additional features should be considered for more comprehensive analysis.